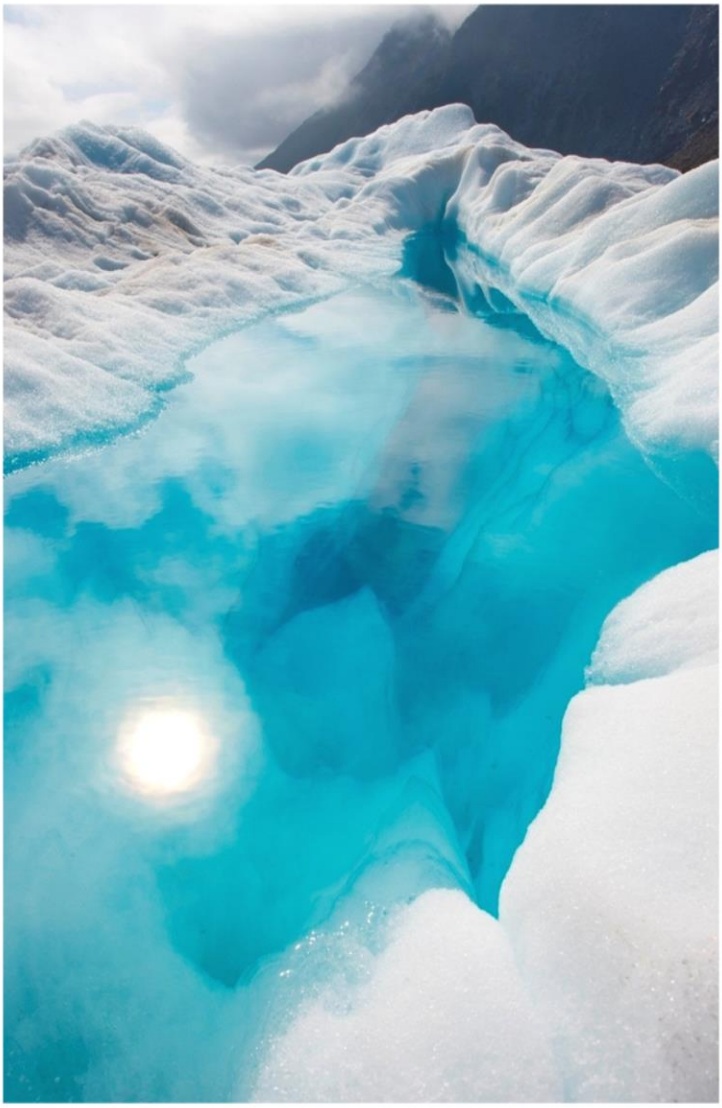


Hardware

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Hardware Table

Component	Brand & Model	Price	Ram	Storage
Router	Mikrotik HEX S RB760IGS	83.50$	256 MB	16MB
Firewall	Fortinet FortiGate 30E	212$	512MB	4GB Flash
Switch 1/2/3	TP-Link TL-SG1024DE (24-Port)	91.69$	N/A	N/A
Servers * 8	HP ProLiant ML30 Gen9	314.99$	32GB	1 x 3.20TB SSD
pc's	ASUS VivoBook 15	399.99$	16GB DDR4	512GB SSD
Laptops	Dell OptiPlex 5070 Micro	450$	8GB up to 16GB	512GB SSD
Cloud	Sync.com	6$	none	1TB

Processor	Frequency	CPU	GPU
880MHz Dual-Core	880 MHz	Qualcomm IPQ4018 (ARM Cortex-A7)	none
FortiSoC2	1.2GHz	Fortinet SoC2 (ARM Cortex-A9)	none
N/A	N/A	none	none
Intel Xeon E-3-1220 v5	3GHz	Intel Xeon E3-1220 v5 (Quad-Core)	none
Intel core i5-9500T	2.2 GHz	Intel Core i5-9500T (6-Core)	intel UHDGraphics 630
Intel core i5 13th gen/ AMD Rayzen 5	~2.5-4.5 CHz	Intel Core i5-13xxx or Ryzen 5	Integrate Graphics
none	none	none	none

Hardware Technical Description

The next section outlines the most important hardware components employed within the design and implementation of our network's big data architecture. Every device was chosen based on its performance, reliability, cost-effectiveness, and support for enterprise-level applications. The details include brand, model, memory, capacity, processing power, and availability of specialized processors like GPUs or DPUs.

Router – Mikrotik HEX S RB760IGS

The Mikrotik HEX S RB760IGS was selected for its compact form factor, high-level routing features, and reasonable cost. Equipped with a dual-core Qualcomm IPQ4018 ARM Cortex-A7 clocked at 880 MHz and 256 MB RAM, this router features top-level performance suitable for small to medium-sized WAN environments. While it doesn't have a GPU or DPU, its internal chipset is optimized for the highest level of throughput for routing and does feature hardware-accelerated encryption for IPsec. Its 5 Gigabit Ethernet and SFP port facilitate versatile uplink schemes, and its price of $83.50 makes it an excellent choice for cost-sensitive network projects *(Mikrotik, 2020)*.

Firewall – Fortinet FortiGate 30E

Security is a concern for any WAN, and the Fortinet FortiGate 30E firewall provides complete network protection. With a FortiSoC2 system-on-chip (ARM Cortex-A9) processor clocked at 1.2 GHz and 512 MB of system memory, this hardware appliance is particularly designed for security operations on the network. It also has an embedded DPU (Data Processing Unit) for dealing with DPI (Deep

Packet Inspection), IPS (Intrusion Prevention System), and antivirus functionality at full wire speed. With 4 GB flash capacity and an embedded FortiOS software stack, this firewall is a next-generation security solution offered in a compact, efficient package for a reasonable price of $212 *(Fortinet, 2020)*.

Switches – TP-Link TL-SG1024DE (x3)

Three TP-Link TL-SG1024DE 24-port Gigabit managed switches were installed to serve the network backbone. Though they lack RAM, memory, processors, and specialized components like GPUs and DPUs, they provide necessary Layer 2+ features which include VLAN support, traffic monitoring, and QoS. The managed switches total 72 Gigabit ports, supporting server interconnects and endpoint devices. With an affordable price ($91.69 each) and being plug-and-play, they are perfect for expandable enterprise-grade LAN segments within the WAN *(TP-Link, 2014)*.

Servers – HP ProLiant ML30 Gen9 (x8)

The WAN infrastructure, consisting of eight HP ProLiant ML30 Gen9 servers, features 32 GB RAM and 3.2 TB SSD, powered by 3 GHz clock speed-based Intel Xeon E3-1220 v5 quad-core processors. The server model is specifically optimized for small and medium-sized enterprise workloads such as database hosting, file storing, application deployment, and big data analytics. The workloads for which the servers are being utilized don't involve any GPU requirements, nor are any discrete DPUs integrated, since most of the operations are carried out through efficient Xeon architecture. With a price tag of $314.99 per server, it is an affordable balance between scalability and performance *(Hewlett Packard Enterprise, 2018a; Hewlett Packard Enterprise, 2018b)*.

PCs – ASUS VivoBook 15

User workstations are equipped with ASUS VivoBook 15 laptops. Each PC features 16 GB of DDR4 RAM and a 512 GB SSD, offering fast boot times and reliable local storage. The Intel Core i5-9500T processor (6-core, 2.2 GHz) ensures smooth multitasking, while integrated Intel UHD Graphics 630 supports basic visual processing needs without requiring a discrete GPU. These machines are ideal for staff accessing applications, performing development tasks, and running basic data queries. At $399.99 each, they offer exceptional performance for office-level computing *(Oxaam, 2025)*.

Laptops – Dell OptiPlex 5070 Micro

The Dell OptiPlex 5070 Micro laptops are used for mobile administrative activities. Equipped with 8 GB (expandable to 16 GB) RAM and a 512 GB SSD, they provide desktop PC-level performance in a small package. Depending on the model, the systems rely on the use of either an Intel Core i5 13th-gen or AMD Ryzen 5 CPU. The two processor series provide high-performance functionality across business applications. Integrated graphics eliminate the necessity for a separate GPU. These systems, costing $450 each, are chosen for their compact size and versatility *(Dell Technologies, 2019)*.

Cloud Storage – Sync.com

For redundancy and scalable off-site backup, WAN architecture is integrated with Sync.com's cloud storage. Though it carries zero hardware, it enables 1 TB secure, encrypted data per subscription at only $6/month. Sync.com is compliant with HIPAA, GDPR, and PIPEDA and is thus apt for dealing with sensitive corporate data. The solution serves as a backup and file-sharing solution for structured and unstructured data sets *(Sync.com, 2025)*.

References

- Mikrotik (2020) *Mikrotik hEX S RB760iGS Router 5X Gigabit Ethernet, SFP, Dual Core 880MHz CPU*. Available at: https://www.amazon.com/Mikrotik-RB760iGS-Router-Gigabit-Ethernet/dp/B0843WLVR4 (Accessed: 1 May 2025).
- Fortinet (2020) *Fortinet FortiGate 30E Network Security Appliance*. Available at: https://www.amazon.com/FORTINET-FC-10-0030E-928-02-12-FortiGate-30E-Application-FortiSandbox/dp/B07BT58BZ8 (Accessed: 1 May 2025).
- TP-Link (2014) *TP-Link TL-SG1024DE 24-Port Gigabit Easy Smart Switch*. Available at: https://www.amazon.com/TP-LINK-TL-SG1024DE-24-Port-Gigabit-Switch/dp/B00CUG8ESM (Accessed: 1 May 2025).
- Hewlett Packard Enterprise (2018) *HP ProLiant ML30 Gen9 Tower Server, Intel Xeon E-2224, 16GB DDR4, 1TB HDD*. Available at: https://www.ebay.com/itm/146553125267 (Accessed: 2 May 2025).
- Dell Technologies (2019) *Dell OptiPlex 5070 Micro, Intel Core i5-9500T, 16GB RAM, 512GB SSD, Windows 10 Pro*. Available at: https://www.ebay.com/itm/226013328264 (Accessed: 2 May 2025).
- Sync.com (2025) *Compare Plans and Pricing*. Available at: https://www.sync.com/pricing/ (Accessed: 3 May 2025).
- Oxaam (2025) *Best budget laptops 2025*. Available at: https://www.oxaam.com/articles/best-budget-laptops-2025/?utm_source=chatgpt.com (Accessed: 16 May 2025).